

Creation Date 06-Sep-2022

Revision Date 12-Oct-2023

Revision Number 2

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product identifier

Product Description: **2.9% CX-100 aziridine crosslinker in butyl acetate**
Cat No. : **630300000**
Molecular Formula **C24 H41 N3 O6**

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use	Laboratory chemicals.
Sector of use	SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites
Product category	PC21 - Laboratory chemicals
Process categories	PROC15 - Use as a laboratory reagent
Environmental release category	ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates)
Uses advised against	No Information available

1.3. Details of the supplier of the safety data sheet

Company

EU entity/business name
Thermo Fisher Scientific
Janssen Pharmaceuticaaan 3a, 2440 Geel,
Belgium

UK entity/business name
Fisher Scientific UK
Bishop Meadow Road,
Loughborough, Leicestershire LE11 5RG,
United Kingdom

Swiss distributor - Fisher Scientific AG
Neuhofstrasse 11, CH 4153 Reinach
Tel: +41 (0) 56 618 41 11
e-mail - infoch@thermofisher.com

E-mail address begel.sdsdesk@thermofisher.com

1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99
CHEMTREC Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:
Tox Info Suisse Emergency Number: **145 (24hr)**
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)
Chemtrec (24h) Toll-Free: 0800 564 402
Chemtrec Local: +41-43 508 20 11 (Zurich)

SECTION 2: HAZARDS IDENTIFICATION

SAFETY DATA SHEET

2.9% CX-100 aziridine crosslinker in butyl acetate

Revision Date 12-Oct-2023

2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

Flammable liquids

Category 3 (H226)

Health hazards

Serious Eye Damage/Eye Irritation

Category 2 (H319)

Skin Sensitization

Category 1 (H317)

Germ Cell Mutagenicity

Category 2 (H341)

Specific target organ toxicity - (single exposure)

Category 3 (H336)

Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

2.2. Label elements



Signal Word

Warning

Hazard Statements

H226 - Flammable liquid and vapor

H317 - May cause an allergic skin reaction

H319 - Causes serious eye irritation

H336 - May cause drowsiness or dizziness

H341 - Suspected of causing genetic defects

EUH066 - Repeated exposure may cause skin dryness or cracking

Precautionary Statements

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P312 - Call a POISON CENTER or doctor if you feel unwell

P280 - Wear protective gloves/protective clothing/eye protection/face protection

2.3. Other hazards

SAFETY DATA SHEET

2.9% CX-100 aziridine crosslinker in butyl acetate

Revision Date 12-Oct-2023

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB)

This product does not contain any known or suspected endocrine disruptors

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.2. Mixtures

Component	CAS No	EC No	Weight %	CLP Classification - Regulation (EC) No 1272/2008
1-Aziridinepropanoic acid, 2-methyl-, 2-ethyl-2-[[3-(2-methyl-1-aziridinyl)-1-oxopropoxy]methyl]-1,3-propanediyl ester	64265-57-2	EEC No. 264-763-3	2.9	Skin Irrit. 2 (H315) Eye Dam. 1 (H318) Skin Sens. 1 (H317) Muta. 2 (H341)
n-Butyl acetate	123-86-4	EEC No. 204-658-1	97.1	Flam. Liq. 3 (H226) STOT SE 3 (H336) EUH066

Full text of Hazard Statements: see section 16

SECTION 4: FIRST AID MEASURES

4.1. Description of first aid measures

General Advice	If symptoms persist, call a physician.
Eye Contact	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.
Skin Contact	Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.
Ingestion	Clean mouth with water and drink afterwards plenty of water.
Inhalation	Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.
Self-Protection of the First Aider	Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

4.2. Most important symptoms and effects, both acute and delayed

May cause allergic skin reaction. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician	Treat symptomatically.
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SECTION 5: FIREFIGHTING MEASURES

5.1. Extinguishing media

SAFETY DATA SHEET

2.9% CX-100 aziridine crosslinker in butyl acetate

Revision Date 12-Oct-2023

Suitable Extinguishing Media

Water mist may be used to cool closed containers. Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. Dry chemical, CO₂, water spray or alcohol-resistant foam.

Extinguishing media which must not be used for safety reasons

No information available.

5.2. Special hazards arising from the substance or mixture

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

Hazardous Combustion Products

None under normal use conditions.

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.

6.2. Environmental precautions

Should not be released into the environment. Do not flush into surface water or sanitary sewer system.

6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges.

Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

7.2. Conditions for safe storage, including any incompatibilities

Keep container tightly closed in a dry and well-ventilated place. Keep away from heat, sparks and flame.

Technical Rules for Hazardous Substances (TRGS) 510

Class 3

ACR63030

SAFETY DATA SHEET

2.9% CX-100 aziridine crosslinker in butyl acetate

Revision Date 12-Oct-2023

Storage Class (LGK) (Germany)

Switzerland - Storage of hazardous substances

Storage class - SC 3

<https://www.kvu.ch/de/themen/stoffe-und-produkte>

<https://www.kvu.ch/fr/themes/substances-et-produits>

<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

7.3. Specific end use(s)

Use in laboratories

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Exposure limits

List source(s): **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

Component	European Union	The United Kingdom	France	Belgium	Spain
n-Butyl acetate		STEL: 200 ppm 15 min STEL: 966 mg/m ³ 15 min TWA: 150 ppm 8 hr TWA: 724 mg/m ³ 8 hr	TWA / VME: 50 ppm (8 heures). restrictive limit TWA / VME: 241 mg/m ³ (8 heures). restrictive limit STEL / VLCT: 150 ppm. restrictive limit STEL / VLCT: 723 mg/m ³ . restrictive limit	TWA: 50 ppm 8 uren TWA: 238 mg/m ³ 8 uren STEL: 150 ppm 15 minuten STEL: 712 mg/m ³ 15 minuten	STEL / VLA-EC: 150 ppm (15 minutos). STEL / VLA-EC: 723 mg/m ³ (15 minutos). TWA / VLA-ED: 50 ppm (8 horas) TWA / VLA-ED: 241 mg/m ³ (8 horas)

Component	Italy	Germany	Portugal	The Netherlands	Finland
n-Butyl acetate	TWA: 241 mg/m ³ 8 ore. Time Weighted Average TWA: 50 ppm 8 ore. Time Weighted Average STEL: 723 mg/m ³ 15 minuti. Short-term STEL: 150 ppm 15 minuti. Short-term	TWA: 62 ppm (8 Stunden). AGW - exposure factor 2 TWA: 300 mg/m ³ (8 Stunden). AGW - exposure factor 2 TWA: 100 ppm (8 Stunden). MAK TWA: 480 mg/m ³ (8 Stunden). MAK Höhepunkt: 200 ppm Höhepunkt: 960 mg/m ³	STEL: 150 ppm 15 minutos STEL: 723 mg/m ³ 15 minutos TWA: 50 ppm 8 horas TWA: 241 mg/m ³ 8 horas	STEL: 723 mg/m ³ 15 minuten TWA: 241 mg/m ³ 8 uren	TWA: 50 ppm 8 tunteina TWA: 240 mg/m ³ 8 tunteina STEL: 150 ppm 15 minuutteina STEL: 725 mg/m ³ 15 minuutteina

Component	Austria	Denmark	Switzerland	Poland	Norway
n-Butyl acetate	MAK-KZGW: 100 ppm 15 Minuten MAK-KZGW: 480 mg/m ³ 15 Minuten MAK-TMW: 50 ppm 8 Stunden MAK-TMW: 241 mg/m ³ 8 Stunden	TWA: 50 ppm 8 timer TWA: 241 mg/m ³ 8 timer STEL: 723 mg/m ³ 15 minutter STEL: 150 ppm 15 minutter	STEL: 150 ppm 15 Minuten STEL: 720 mg/m ³ 15 Minuten TWA: 50 ppm 8 Stunden TWA: 240 mg/m ³ 8 Stunden	STEL: 720 mg/m ³ 15 minutach TWA: 240 mg/m ³ 8 godzinach	TWA: 241 mg/m ³ 8 timer TWA: 50 ppm 8 timer STEL: 723 mg/m ³ 15 minutter. value from the regulation STEL: 150 ppm 15 minutter. value from the regulation

Component	Bulgaria	Croatia	Ireland	Cyprus	Czech Republic
n-Butyl acetate	TWA: 241 mg/m ³ TWA: 50 ppm STEL : 723 mg/m ³ STEL : 150 ppm	TWA-GVI: 50 ppm 8 satima. TWA-GVI: 241 mg/m ³ 8 satima. STEL-KGVI: 150 ppm 15 minutama. STEL-KGVI: 723 mg/m ³	STEL: 150 ppm 15 min STEL: 723 mg/m ³ 15 min	STEL: 723 mg/m ³ STEL: 150 ppm TWA: 241 mg/m ³ TWA: 50 ppm	TWA: 241 mg/m ³ 8 hodinách. Ceiling: 723 mg/m ³

SAFETY DATA SHEET

2.9% CX-100 aziridine crosslinker in butyl acetate

Revision Date 12-Oct-2023

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Component	Estonia	Gibraltar	Greece	Hungary	Iceland
n-Butyl acetate	TWA: 241 mg/m ³ 8 tundides. TWA: 50 ppm 8 tundides. STEL: 723 mg/m ³ 15 minutites. STEL: 150 ppm 15 minutites.		STEL: 150 ppm STEL: 723 mg/m ³ TWA: 50 ppm TWA: 241 mg/m ³	STEL: 723 mg/m ³ 15 percekben. CK TWA: 241 mg/m ³ 8 órában. AK	STEL: 150 ppm STEL: 723 mg/m ³ TWA: 50 ppm 8 klukkustundum. TWA: 241 mg/m ³ 8 klukkustundum. Ceiling: 300 ppm Ceiling: 1400 mg/m ³

Component	Latvia	Lithuania	Luxembourg	Malta	Romania
n-Butyl acetate	STEL: 723 mg/m ³ STEL: 150 ppm TWA: 241 mg/m ³ TWA: 50 ppm	TWA: 241 mg/m ³ IPRD TWA: 50 ppm IPRD STEL: 723 mg/m ³ STEL: 150 ppm	TWA: 241 mg/m ³ 8 Stunden TWA: 50 ppm 8 Stunden STEL: 723 mg/m ³ 15 Minuten STEL: 150 ppm 15 Minuten	TWA: 50 ppm TWA: 214 mg/m ³ STEL: 150 ppm 15 minuti STEL: 723 mg/m ³ 15 minuti	TWA: 150 ppm 8 ore TWA: 715 mg/m ³ 8 ore STEL: 200 ppm 15 minute STEL: 950 mg/m ³ 15 minute

Component	Russia	Slovak Republic	Slovenia	Sweden	Turkey
n-Butyl acetate	TWA: 50 mg/m ³ 0429 MAC: 200 mg/m ³	Ceiling: 700 mg/m ³ TWA: 100 ppm TWA: 500 mg/m ³	TWA: 241 mg/m ³ 8 urah TWA: 50 ppm 8 urah STEL: 150 ppm 15 minutah STEL: 723 mg/m ³ 15 minutah	Binding STEL: 150 ppm 15 minuter Binding STEL: 723 mg/m ³ 15 minuter TLV: 50 ppm 8 timmar. NGV TLV: 241 mg/m ³ 8 timmar. NGV	

Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS70 General methods for sampling airborne gases and vapours

MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography

MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

No information available

Predicted No Effect Concentration (PNEC)

See values below.

Component	Fresh water	Fresh water sediment	Water Intermittent	Microorganisms in sewage treatment	Soil (Agriculture)
n-Butyl acetate 123-86-4 (97.1)	PNEC = 0.18mg/L	PNEC = 0.981mg/kg	PNEC = 0.36mg/L	PNEC = 35.6mg/L	PNEC = 0.0903mg/kg soil

SAFETY DATA SHEET

2.9% CX-100 aziridine crosslinker in butyl acetate

Revision Date 12-Oct-2023

		sediment dw			dw
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Component	Marine water	Marine water sediment	Marine water Intermittent	Food chain	Air
n-Butyl acetate 123-86-4 (97.1)	PNEC = 0.018mg/L	PNEC = 0.0981mg/kg sediment dw			

8.2. Exposure controls

Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting/equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Personal protective equipment

Eye Protection Goggles (European standard - EN 166)

Hand Protection Protective gloves

Glove material	Breakthrough time	Glove thickness	EU standard	Glove comments
Nitrile rubber	See manufacturers	-	EN 374	(minimum requirement)
Neoprene	recommendations			
Natural rubber				
PVC				

Skin and body protection Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

Respiratory Protection When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

Large scale/emergency use Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced
Recommended Filter type: Organic gases and vapours filter Type A Brown conforming to EN14387

Small scale/Laboratory use Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.
Recommended half mask:- Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141
When RPE is used a face piece Fit Test should be conducted

Environmental exposure controls Prevent product from entering drains.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Physical State Liquid

Appearance Light yellow

SAFETY DATA SHEET

2.9% CX-100 aziridine crosslinker in butyl acetate

Revision Date 12-Oct-2023

Odor	Characteristic	
Odor Threshold	No data available	
Melting Point/Range	-76 °C / -104.8 °F	
Softening Point	No data available	
Boiling Point/Range	124 °C / 255.2 °F	
Flammability (liquid)	Flammable	On basis of test data
Flammability (solid,gas)	Not applicable	Liquid
Explosion Limits	Lower 3 vol % Upper 10.4 vol %	
Flash Point	27 °C / 80.6 °F	Method - No information available
Autoignition Temperature	No data available	
Decomposition Temperature	No data available	
pH	5 @ 20°C	
Viscosity	No data available	
Water Solubility	No information available	
Solubility in other solvents	No information available	
Partition Coefficient (n-octanol/water)		
Component	log Pow	
n-Butyl acetate	2.3	
Vapor Pressure	10.7 hPa @ 20°C	
Density / Specific Gravity	0.88	
Bulk Density	Not applicable	Liquid
Vapor Density	No data available	(Air = 1.0)
Particle characteristics	Not applicable (liquid)	

9.2. Other information

Molecular Formula	C24 H41 N3 O6
Molecular Weight	467.6

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity

None known, based on information available

10.2. Chemical stability

Stable under normal conditions.

10.3. Possibility of hazardous reactions

Hazardous Polymerization	No information available.
Hazardous Reactions	None under normal processing.

10.4. Conditions to avoid

Keep away from open flames, hot surfaces and sources of ignition.

10.5. Incompatible materials

None known.

10.6. Hazardous decomposition products

None under normal use conditions.

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

SAFETY DATA SHEET

2.9% CX-100 aziridine crosslinker in butyl acetate

Revision Date 12-Oct-2023

Product Information

(a) acute toxicity;

Oral

Based on available data, the classification criteria are not met

Dermal

Based on available data, the classification criteria are not met

Inhalation

Based on available data, the classification criteria are not met

Toxicology data for the components

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
n-Butyl acetate	LD50 = 10768 mg/kg (Rat)	LD50 > 17600 mg/kg (Rabbit)	LC50 = 0.74 mg/L (Rat) 4 h

(b) skin corrosion/irritation;

Based on available data, the classification criteria are not met

(c) serious eye damage/irritation;

Category 2

(d) respiratory or skin sensitization;

Respiratory

Based on available data, the classification criteria are not met

Skin

Category 1

No information available

(e) germ cell mutagenicity;

Category 2

(f) carcinogenicity;

Based on available data, the classification criteria are not met

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity;

Based on available data, the classification criteria are not met

(h) STOT-single exposure;

Category 3

Results / Target organs

Central nervous system (CNS).

(i) STOT-repeated exposure;

Based on available data, the classification criteria are not met

Target Organs

None known.

(j) aspiration hazard;

Based on available data, the classification criteria are not met

Symptoms / effects, both acute and delayed

Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

11.2. Information on other hazards

Endocrine Disrupting Properties

Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

SECTION 12: ECOLOGICAL INFORMATION

12.1. Toxicity

Ecotoxicity effects

Contains a substance which is: Harmful to aquatic organisms. The product contains

SAFETY DATA SHEET

2.9% CX-100 aziridine crosslinker in butyl acetate

Revision Date 12-Oct-2023

following substances which are hazardous for the environment.

Component	Freshwater Fish	Water Flea	Freshwater Algae
n-Butyl acetate	Lepomis macrochirus: LC50: 100 mg/L/96h Pimephales promelas: LC50:17-19 mg/L/96h		EC50: = 674.7 mg/L, 72h (Desmodesmus subspicatus)

Component	Microtox	M-Factor
n-Butyl acetate	EC50 = 70.0 mg/L 5 min EC50 = 82.2 mg/L 15 min EC50 = 959 mg/L 18 h EC50 = 98.9 mg/L 30 min	

12.2. Persistence and degradability

Persistence

Degradation in sewage treatment plant

Persistence is unlikely.

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

12.3. Bioaccumulative potential

Bioaccumulation is unlikely

Component	log Pow	Bioconcentration factor (BCF)
n-Butyl acetate	2.3	No data available

12.4. Mobility in soil

No information available

12.5. Results of PBT and vPvB assessment

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB).

12.6. Endocrine disrupting properties

Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

12.7. Other adverse effects

Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance
This product does not contain any known or suspected substance

SECTION 13: DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

Contaminated Packaging

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

European Waste Catalogue (EWC)

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

Other Information

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations.

SAFETY DATA SHEET

2.9% CX-100 aziridine crosslinker in butyl acetate

Revision Date 12-Oct-2023

Switzerland - Waste Ordinance Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

SECTION 14: TRANSPORT INFORMATION

IMDG/IMO

14.1. UN number UN1123
14.2. UN proper shipping name BUTYL ACETATES
14.3. Transport hazard class(es) 3
14.4. Packing group III

ADR

14.1. UN number UN1123
14.2. UN proper shipping name BUTYL ACETATES
14.3. Transport hazard class(es) 3
14.4. Packing group III

IATA

14.1. UN number UN1123
14.2. UN proper shipping name BUTYL ACETATES
14.3. Transport hazard class(es) 3
14.4. Packing group III

14.5. Environmental hazards No hazards identified
14.6. Special precautions for user No special precautions required.
14.7. Maritime transport in bulk according to IMO instruments Not applicable, packaged goods

SECTION 15: REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

Component	CAS No	EINECS	ELINCS	NLP	IECSC	TCSI	KECL	ENCS	ISHL
1-Aziridinepropanoic acid, 2-methyl-, 2-ethyl-2-[[3-(2-methyl-1-aziridinyl)-1-oxopropoxy]methyl]-1,3-propanediyl ester	64265-57-2	264-763-3	-	-	X	X	KE-23443	-	X
n-Butyl acetate	123-86-4	204-658-1	-	-	X	X	KE-04179	X	X

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	DSL	NDSL	AICS	NZIoC	PICCS
1-Aziridinepropanoic acid, 2-methyl-, 2-ethyl-2-[[3-(2-methyl-1-aziridinyl)-1-oxopropoxy]methyl]-1,3-propanediyl ester	64265-57-2	X	ACTIVE	X	-	X	X	-

SAFETY DATA SHEET

2.9% CX-100 aziridine crosslinker in butyl acetate

Revision Date 12-Oct-2023

ediyl ester								
n-Butyl acetate	123-86-4	X	ACTIVE	X	-	X	X	X

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

Authorisation/Restrictions according to EU REACH

Not applicable

Component	CAS No	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
1-Aziridinepropanoic acid, 2-methyl-, 2-ethyl-2-[[3-(2-methyl-1-aziridinyl)-1-oxopropoxy]methyl]-1,3-propanediyl ester	64265-57-2	-	-	-
n-Butyl acetate	123-86-4	-	-	-

Seveso III Directive (2012/18/EC)

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements
1-Aziridinepropanoic acid, 2-methyl-, 2-ethyl-2-[[3-(2-methyl-1-aziridinyl)-1-oxopropoxy]methyl]-1,3-propanediyl ester	64265-57-2	Not applicable	Not applicable
n-Butyl acetate	123-86-4	Not applicable	Not applicable

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

WGK Classification

Water endangering class = 1 (self classification)

Component	Germany - Water Classification (AwSV)	Germany - TA-Luft Class
n-Butyl acetate	WGK1	

Component	France - INRS (Tables of occupational diseases)
n-Butyl acetate	Tableaux des maladies professionnelles (TMP) - RG 84

Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

SAFETY DATA SHEET

2.9% CX-100 aziridine crosslinker in butyl acetate

Revision Date 12-Oct-2023

Component	Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81)	Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC)	Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure
n-Butyl acetate 123-86-4 (97.1)		Group I	

15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

SECTION 16: OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3

H317 - May cause an allergic skin reaction
H319 - Causes serious eye irritation
H336 - May cause drowsiness or dizziness
H341 - Suspected of causing genetic defects
EUH066 - Repeated exposure may cause skin dryness or cracking
H226 - Flammable liquid and vapor
H315 - Causes skin irritation
H318 - Causes serious eye damage

Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

DNEL - Derived No Effect Level

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

NOEC - No Observed Effect Concentration

PBT - Persistent, Bioaccumulative, Toxic

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer
Predicted No Effect Concentration (PNEC)

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadviser - LOLI, Merck index, RTECS

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (volatile organic compound)

Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

Physical hazards On basis of test data

Health Hazards Calculation method

Environmental hazards Calculation method

Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

SAFETY DATA SHEET

2.9% CX-100 aziridine crosslinker in butyl acetate

Revision Date 12-Oct-2023

First aid for chemical exposure, including the use of eye wash and safety showers.

Creation Date	06-Sep-2022
Revision Date	12-Oct-2023
Revision Summary	SDS sections updated.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,
Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and
Preparations).**

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of Safety Data Sheet